

Fixed-margin prediction of cell-level RNA–surface-protein dependence across cohorts

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Abstract

Background: Linked single-cell assays resolve cross-modal dependence because they measure molecular states in the same cell. A recipient cohort may retain the marginal state counts of both assays without preserving cell-level pairing. Its joint table is then unobserved and cannot be recovered from the margins alone.

Results: We formulate prediction from paired source donors and unpaired recipient margins. Penalized exact conditional likelihood estimates donor-specific log odds and a shared population interaction; the predicted table is the conditional mean at the recipient margins. This table uniquely minimizes expected multinomial deviance under the conditional model, and a finite-sample bound separates source estimation error from recipient association drift. Using RNA detection and within-donor median-ranked surface-protein states in 512 cells per donor, a model fitted on 36 Cambridge donors achieved mean deviance per cell 0.01220 in 56 Newcastle donors. Paule–Mandel random-effects transfer of the signed-root deviance statistic reached 0.01478, a 17.46% reduction for the conditional estimator (paired-donor 95% confidence interval for the loss difference, -0.00413 to -0.00080; 50/56 donors favorable). Mean deviance after source linkage was destroyed was 0.02274. In post hoc fitted-baseline analyses, the estimator reduced deviance by 6.18% relative to the common-effect conditional estimate (95% confidence interval, 3.84% to 8.47%). Across 128 ground-truth simulations, conditioning absorbed recipient margin shift, and donor shrinkage improved prediction when source interactions were heterogeneous.

Conclusions: Fixed-margin prediction reconstructs an unobserved cross-modal joint table from linked source donors and recipient aggregate counts. The held-site result supports transfer of RNA–surface-protein dependence for the studied marker panel, and the simulations identify when conditioning and donor shrinkage improve prediction.

Keywords: single-cell multi-omics; conditional likelihood; log-linear association; transfer learning; CITE-seq; reproducibility

1 Background

Linked single-cell assays measure RNA, chromatin, proteins, molecular age, and lineage in the same cell or experimentally related unit [1–6]. Joint latent-variable models integrate paired measurements or predict a missing modality. We consider a different information boundary: source donors have paired

measurements, whereas a recipient provides only the aggregate state counts for each assay. The target is the recipient joint table that cell-level pairing would have revealed.

Marginal frequencies change with cell composition, activation, sequencing depth, and assay scale. Copying a source table therefore transfers abundance together with dependence. Signed Pearson

and likelihood-ratio statistics summarize departure from independence, but their scale and attainable range depend on the margins. A statistic pooled across donors can consequently change meaning at recipient margins.

Log-linear interactions isolate association in contingency tables [7–10]. Conditional likelihood removes row and column nuisance parameters; hierarchical conditional and random-effects log-linear models pool odds ratios across observed strata [11–14]. Fixed-margin tables can be generated from specified log-linear dependence [15], and marginal distributions can be completed by odds-ratio coordinates in variation-independent parameterizations [16]. Categorical data fusion also studies joint distributions that are not directly observed, using proxies or other auxiliary identifying information [17]. Copula and differential-association methods estimate dependence from observed single-cell or multi-omic pairings [18–21].

Our contribution is a predictive formulation for source-paired, recipient-unpaired studies. The method learns a transferable dependence coordinate from stratified source tables, combines it with each recipient’s observed margins, and returns the conditional mean joint table. This action is optimal under expected multinomial deviance for the fitted conditional law. A risk bound separates error in the transported source interaction from irreducible recipient drift. Unlike observed-table meta-analysis, prediction does not use recipient pairing; unlike categorical data fusion, it does not require recipient-level proxy variables beyond the two margins. A prespecified, pairing-blinded CITE-seq experiment and ground-truth simulations test the estimator against matched classical procedures and margin-preserving link destruction.

2 Methods

2.1 Association coordinates

For ordered marker pair e , let P_e be a positive joint-probability table for linked states U and V . Orthonormal contrast matrices B_U and B_V remove the row and column components of its log table. We call the resulting centered log-linear interaction the *coupling field*:

$$\Theta(P_e) = B_U^T \log(P_e) B_V \quad (1)$$

where the logarithm is elementwise. This field contains all log-linear interaction degrees of freedom, is invariant to positive row or column tilts, and vanishes exactly at independence.

For a binary table, Eq. (1) is one-half of the log odds ratio,

$$\Theta(P_e) = \frac{1}{2} \log \frac{P_{e,00} P_{e,11}}{P_{e,01} P_{e,10}} = \frac{\theta_e}{2}. \quad (2)$$

Proposition S1 in Additional file 1 proves these properties.

2.2 Hierarchical conditional estimator

Let A_{de} be the upper-left count in the binary table for donor d and marker pair e , and let m_{de} denote its margins. Conditional on these margins, the table has one free count. Its exact distribution is

$$p(a \mid m_{de}, \theta) = \frac{w_{de}(a) \exp(\theta a)}{\sum_{b \in \mathcal{A}_{de}} w_{de}(b) \exp(\theta b)}, \quad (3)$$

where $w_{de}(a)$ is the central-hypergeometric combinatorial weight. The distribution depends on the log odds θ but not on row or column effects.

We write the donor-specific log odds as a population effect μ_e plus a donor deviation δ_{de} . The estimator minimizes

$$\begin{aligned} \mathcal{L}(\mu, \delta) = & -\frac{1}{D} \sum_{d,e} \log p(A_{de} \mid m_{de}, \mu_e + \delta_{de}) \\ & + \frac{\gamma}{2} \sum_{d,e} \delta_{de}^2 + \frac{\rho}{2} \|\mu\|_2^2. \end{aligned} \quad (4)$$

The first term is the donor-normalized negative exact conditional log likelihood summed over marker pairs. The panel objective is composite because marker pairs from one donor reuse cells. The parameter γ shrinks donor deviations toward the population interaction, and ρ penalizes the population interaction. Penalty scaling, optimization, and convergence criteria are given in Additional file 1.

Conditioning a Poisson log-linear model on both margins removes every row and column parameter and yields Eq. (3). Positive donor and population penalties give Eq. (4) a unique finite minimizer; Proposition S2 in Additional file 1 gives the derivation and proof.

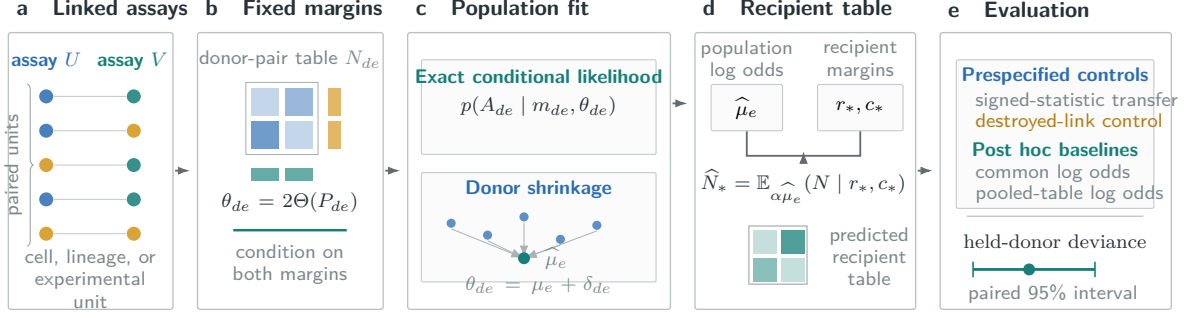


Fig. 1 Conditional prediction of pairwise dependence. Linked source measurements define donor-level joint tables. Conditioning on both margins isolates log-linear dependence, and donor shrinkage estimates the population component shared across donors. Combining that estimate with recipient margins reconstructs the missing recipient table. Held comparisons use random-effects signed-statistic transfer and margin-preserving link destruction; fitted analyses compare common-effect conditional and pooled-table estimates.

For recipient q , the prediction is the fixed-margin expected table at transported population log odds,

$$\hat{N}_{qe} = \mathbb{E}_{\alpha \hat{\mu}_e} [N_{qe} \mid m_{qe}], \quad (5)$$

where the development design selects α from a prespecified study grid. The binary cohort studies used $\{0.5, 0.75, 1, 1.25\}$, and the longitudinal calibration also included zero. This table preserves the recipient margins without a pseudocount or post-fit projection.

Proposition S4 in Additional file 1 proves that the conditional mean in Eq. (5) uniquely minimizes expected multinomial deviance among tables with the recipient margins. Proposition S3 bounds expected excess conditional log loss by fixed-margin Fisher information times the sum of squared source transport error and recipient drift variance. Source data can reduce the first term, whereas recipient drift determines the residual floor. We assessed held reconstructed tables separately by multinomial deviance.

2.3 Classical comparisons and controls

The matched statistic comparator transfers either the signed root Pearson statistic or the signed root likelihood-ratio deviance from the row-plus-column independence model. We normalized statistics by table size, pooled them across source donors, rescaled them for each recipient, and inverted them at its margins. The Stephenson protocol used Paule–Mandel random-effects pooling [22]. Each

study fixed its pooling rule in advance; development data selected the statistic, any evaluated null centering, and the transport multiplier. Both methods received the same recipient margins without recipient pairings.

Two fitted interaction baselines compared alternative pooling rules. The common-effect conditional maximum-likelihood estimate fitted one unpenalized log odds per marker pair across donor strata. The pooled-table baseline used the sample log odds ratio of the donor-pooled table. In the post hoc comparison, both coefficients were reconstructed by fixed-margin conditional expectation with a development-selected transport multiplier. We report the second method as pooled-table log odds with conditional reconstruction.

The destroyed-link control permuted complete protein profiles within each donor before joint tables were formed. This preserved RNA and protein margins and dependence within the protein panel but removed cell-level RNA-protein linkage. We refitted the selected conditional estimator to the permuted tables. Fixed-margin independence provided a further control.

For observed table T , positive prediction $\hat{T}$, and table total n , the loss is multinomial deviance per cell,

$$D(T, \hat{T}) = \frac{2}{n} \sum_{i,j: T_{ij} > 0} T_{ij} \log \frac{T_{ij}}{\hat{T}_{ij}}. \quad (6)$$

Entity losses were averaged within each evaluation unit. Confirmatory means weighted physical donors equally. The 95% intervals used 20,000

paired bootstrap draws at the donor or prespecified batch-block level, conditional on the source fit, selected configuration, and deterministic cell sample. Development intervals resampled the displayed samples or batches.

2.4 Ground-truth simulation

We simulated 128 replicates from the exact fixed-margin model with known population log odds, 12 source donors, 12 recipient donors, 16 marker pairs, and 128 observations per table. Source donor effects were either homogeneous or normally heterogeneous with standard deviation 1.2. Recipient margins either followed the source distribution or underwent a strong shift. Hyperparameters and transport multipliers were fixed across all four scenarios. We compared the hierarchical estimator with the common-effect conditional estimate and raw and null-centered signed-root deviance-statistic transfer. Paired bootstrap intervals resampled simulation replicates in 20,000 draws.

2.5 Prespecified held evaluation

The Stephenson analysis plan fixed eligibility, donor allocation, cell selection, state definitions, marker panels, tuning grids, comparators, loss, uncertainty, and success criteria before the Newcastle pairing outcomes were observed. The Cambridge pilot selected the final configuration from these grids, after which all Cambridge development donors supplied the final fit. Predictions used Newcastle margins alone and were recorded before RNA-protein pairing was restored for evaluation.

Confirmation required at least 5% lower mean deviance than both the selected statistic-transfer and destroyed-link controls, a paired-bootstrap upper endpoint below zero, and a one-sided sign-test $P \leq 0.025$. Model selection ended before the Newcastle pairing outcomes were observed.

2.6 Datasets and state construction

Stephenson et al. profiled PBMC RNA and surface proteins with a common TotalSeq-C panel across Cambridge and Newcastle [23]. The split used 12 Cambridge calibration donors, 24 Cambridge pilot donors, and 56 Newcastle donors for the held-site test. Nine cognate RNA and protein markers defined 81 ordered pairs.

Each donor contributed 512 cells selected by a deterministic rule that did not access RNA or ADT counts. RNA state recorded raw detection. Deterministic within-donor protein ranks assigned 256 low and 256 high cells, fixing every protein margin before evaluation.

3 Results

3.1 Association transferred across recruitment sites

Across 56 Newcastle donors, the hierarchical conditional estimator reached mean deviance 0.01220, compared with 0.01478 for signed-root deviance-statistic transfer. The corresponding 17.46% reduction had a paired-donor loss-difference interval of -0.00413 to -0.00080. The conditional estimate was better in 50/56 donors ($P = 5.09 \times 10^{-10}$, one-sided sign test). All 36 Cambridge development donors supplied the final source fit after pilot selection.

The estimator also reduced deviance by 46.36% relative to the destroyed-link fit of 0.02274 (paired-donor interval -0.01323 to -0.00749; 51/56 donors favorable). In post hoc fitted-baseline analyses, it improved on the common-effect conditional estimate by 6.18% (relative-reduction interval 3.84% to 8.47%) and on pooled-table log odds with conditional reconstruction by 3.39% (1.42% to 5.37%).

3.2 Conditioning and donor shrinkage separated distinct errors

The factorial simulation separated donor heterogeneity from margin shift (Fig. 2). With heterogeneous donor effects and source-like margins, the hierarchical estimator reached deviance 0.01493, compared with 0.01521 for the common-effect conditional estimate. The reduction was 1.85% (95% interval 1.52% to 2.18%). After the recipient margins shifted, conditional transport reduced deviance by 65.49% (63.02% to 67.72%) relative to raw signed-root deviance-statistic transfer.

Under homogeneous donor effects and source-like margins, common-effect conditional deviance was 0.008660, compared with 0.008752 for the hierarchical estimator, a shrinkage cost of 1.06%. After margin shift, the two conditional estimators remained close (0.008640 and 0.008674),

Table 1 Held-site performance against matched controls and fitted interaction baselines.

Comparator	Analysis role	Comparator loss	Reduction	Difference 95% CI	Favorable donors
Signed-root deviance transfer	prespecified	0.01478	17.46%	-0.00413 to -0.00080	50/56
Destroyed source linkage	prespecified	0.02274	46.36%	-0.01323 to -0.00749	51/56
Common-effect conditional estimate	fitted baseline	0.01300	6.18%	-0.00121 to -0.00046	46/56
Pooled-table log odds	fitted baseline	0.01263	3.39%	-0.00075 to -0.00017	38/56

The hierarchical conditional loss was 0.01220. Differences are hierarchical minus comparator deviance per cell. Intervals use 20,000 paired donor-bootstrap draws. Fitted-baseline comparisons were specified after the held result and are descriptive.

whereas statistic-transfer deviance rose to 0.02943. Fixed-margin reconstruction therefore absorbed the marginal shift in this model; donor shrinkage contributed when source associations varied across donors.

Both risk bounds held throughout the finite-support stress grid. Under unbiased transport, risk increased monotonically with recipient drift in every margin family (Additional file 1).

4 Discussion

Recipient margins and source heterogeneity govern different parts of the prediction. Fixed-margin reconstruction absorbed abundance shifts in simulation and transferred RNA–surface-protein dependence from Cambridge to Newcastle without copying source composition. Donor shrinkage improved on common-effect conditional estimation at the held site and under simulated heterogeneity. Its advantage disappeared when simulated source effects were homogeneous, locating the gain in donor pooling rather than in the interaction coordinate alone.

Classical stratified odds-ratio models estimate association among observed tables; fixed-margin construction combines known margins with specified dependence. We join these operations in a predictive study design whose recipient pairing is absent at fit time. The prespecified Paule–Mandel statistic transfer uses the same linked source tables and recipient margins, so its advantage cannot arise from extra observations. Models such as totalVI and MultiVI solve missing-modality problems from paired recipient observations or cell-level covariates and cannot be fitted under this protocol without changing the task. The common-effect conditional

estimate is the closest fitted classical comparison under the same information boundary; hierarchical pooling reduced its held-site deviance by 6.18%.

5 Limitations

The held confirmation is an observational RNA–surface-protein study and does not establish perturbation causality, molecular binding, or a regulatory mechanism. It uses binary states, 512 cells per donor, and nine RNA and nine protein markers. Within-donor median ranks fix each recipient protein margin at 256/256. The held experiment therefore evaluates varying RNA prevalence and site-dependent association at one protein discretization; it does not test shifts in protein prevalence, natural protein thresholds, other quantiles, or other cell budgets. The 81 pairwise tables within a donor reuse cells and marker vectors. Broader panels, technologies, tissues, state resolutions, and marker sets require independent replication.

The held-site selection assigned no weight to graph regularization, so the empirical claim concerns hierarchical conditional transfer without graph or hypergraph smoothing. The prespecified Paule–Mandel comparator pools signed statistics rather than log odds; a separate random-effects log-linear comparator was not fitted for Stephenson. The fitted-interaction comparisons were specified after the held result and are descriptive.

The Stephenson sign test is study specific and does not correct familywise error across the preceding search of public datasets. Its confirmatory interpretation therefore applies to the prespecified Stephenson experiment, not to the full development program. Earlier datasets were screened for availability, matched features, sample support, and

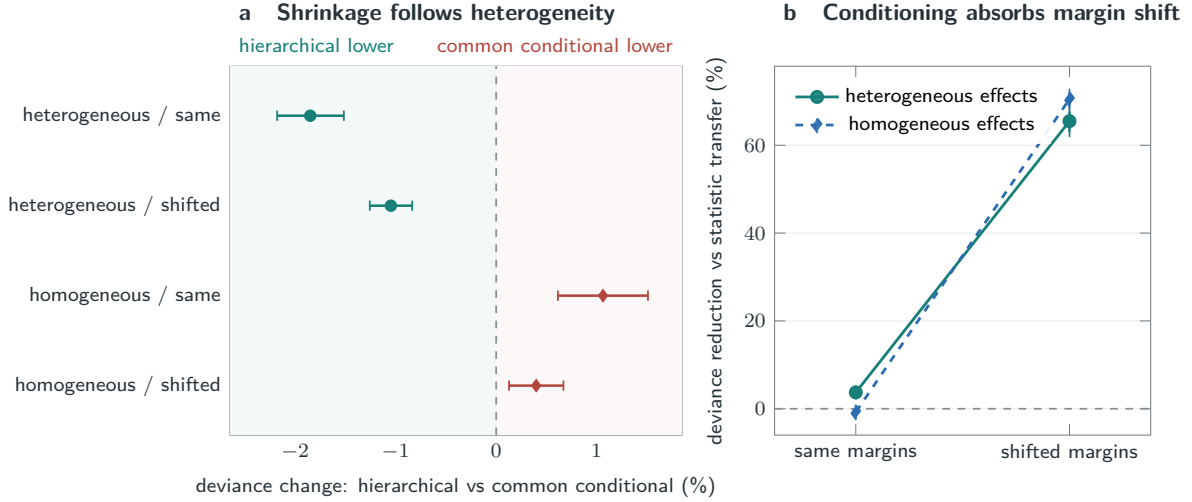


Fig. 2 Ground-truth estimator behavior. **a**, relative change in recipient deviance for the hierarchical estimator against the common-effect conditional estimate. Negative values favor hierarchical shrinkage. **b**, reduction in deviance relative to raw signed-root deviance-statistic transfer before and after recipient-margin shift. Bars are 95% paired simulation-replicate bootstrap intervals across 128 replicates.

development performance. Table 2 and Additional file 1 report the complete sequence, including unsuccessful transfers and a cohort analysis corrected after outcome access.

Table 2 Position of the held-site experiment in the public binary-study sequence.

Last stage reached	Studies	Recipient pairing
Pre-outcome held-site test	1	observed after predictions were fixed
Post-access corrected replication	1	observed before the correction
Calibration or pilot stop	2	not observed
Source or eligibility stop	2	not observed

Conditioning on random margins can discard finite-sample information about an odds ratio under multinomial or independent-binomial sampling. The lost fraction converges to zero under standard large-sample conditions [24], but we did not quantify that loss for these sparse marker tables. Our recipient information boundary makes the margins observed constraints; it does not imply that conditioning is universally more efficient than

an unconditional model when full recipient tables are available.

The exploratory multistate extension depends on discretization, a pseudocount, and finite permutation centering. Spatial applications require linked observations because unpaired regional summaries do not supply the joint tables required by the estimand.

6 Conclusions

Fixed-margin prediction reconstructs cell-level cross-modal dependence from linked source donors and recipient aggregate counts. In the prespecified held-site experiment, hierarchical conditional estimation improved on matched random-effects statistic transfer and destroyed source linkage. Recipient margins address abundance shift, and source heterogeneity determines the value of donor shrinkage.

Abbreviations

ADT: antibody-derived tag; PBMC: peripheral blood mononuclear cell; RNA: ribonucleic acid.

Declarations

Ethics approval and consent to participate

This study used only public, de-identified data and involved no new participants, specimens, interventions, or identifiable records; the original studies report their ethics approvals and consent procedures.

Consent for publication

Not applicable.

Availability of data and materials

All source datasets are public through the accessions listed in Additional file 1 and retain their original licenses. Analysis plans, MIT-licensed software, cohort and simulation results, and study records are available in the v2.0.3 release at <https://github.com/sushaan-k/coupling-fields-benchmark/releases/tag/coupling-fields-v2.0.3-public-benchmark>. Raw matrices are not redistributed.

Competing interests

Competing-interest information is withheld in the blinded manuscript.

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Authors' contributions

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Additional file

Additional file 1 (PDF). *Mathematical results and public benchmark details*. Proofs, algorithms, dataset provenance, study status, comparator definitions, sensitivity analyses, and reproducibility information.

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